CHEAT SHEET — DEMO INPUTS & FLOWS FOR ALL DATASETS

Project: Intelligent Secure Predictive Caching System

📌 Part 1 — Dataset Switching

Use this from the dropdown in your UI.

Dataset	Use For
Healthcare	Patient records, scans, labs
Banking	Transactions, KYC, balance
E-commerce	Products, orders, carts

Switching dataset triggers:
✓ Predictor reload
✓ Domain context switch
✓ Independent cache patterns

📌 Part 2 — Healthcare Dataset — DEMO INPUTS

⭐ Common Inputs

Input	Meaning
HLC_PRESC_101	Prescription
HLC_SCAN_205	Scan/MRI/CT
HLC_LAB_310	Lab test
HLC_RECORD_999	General health record
HLC_VISIT_450	Visit/OPD record

🎯 DEMO FLOW (Best Sequence)

🔹 ***Case 1: Cache Miss → Hit***

1. Input → HLC_PRESC_101 → ❌ Miss
2. Input → HLC_PRESC_101 → ✅ Hit

🔹 ***Case 2: Predictor Preloading***

3. Input → HLC_SCAN_205 → ❌ Miss
   → Predictor preloads: HLC_SCAN_206, HLC_SCAN_207
4. Input → HLC_SCAN_206 → ✅ Hit (Predicted!)

🔹 *Case 3: Random → Miss + Fallback*

5. Input → XYZ_123 → ❌ Miss (fallback to global popularity)

📌 Part 3 — Banking Dataset — DEMO INPUTS

⭐ Common Banking Inputs

Input	Meaning
BNK_TXN_5501	Transaction
BNK_BAL_202	Balance check
BNK_KYC_309	KYC document
BNK_CREDIT_777	Credit statement
BNK_LOAN_450	Loan info

🎯 DEMO FLOW

🔹 *Case 1: Miss → Hit*

1. BNK_TXN_5501 → ❌ Miss
2. BNK_TXN_5501 → ✅ Hit

🔹 *Case 2: Predictor Preloading*

3. BNK_BAL_202 → ❌ Miss → Preloads BNK_TXN_5502
4. BNK_TXN_5502 → ✅ Hit (Predicted!)

🔹 *Case 3: Rare Query*

5. BNK_LOAN_450 → Miss → Cached → Hit next time

📌 Part 4 — E-commerce Dataset — DEMO INPUTS

⭐ Common Ecommerce Inputs

Input	Meaning
ECM_PROD_1101	Product page
ECM_CART_770	Cart info
ECM_ORD_2202	Order history
ECM_REC_450	Recommendation

Input	Meaning
ECM_WISHLIST_990	Wishlist

🎯 DEMO FLOW

🔹 *Case 1: Miss → Hit*

1. ECM_PROD_1101 → ❌ Miss
2. ECM_PROD_1101 → ✅ Hit

🔹 *Case 2: Predictor Preloading*

3. ECM_ORD_2202 → ❌ Miss → Preloads ECM_ORD_2203
4. ECM_ORD_2203 → ✅ Hit (Predicted)

🔹 *Case 3: Fallback Case*

5. Input → NOT_EXISTING_001 → ❌ Miss → Preloads global top

📌 Part 5 — Edge Case Scenarios to Show

✓ Completely New Query

- UNKNOWN_ID_999
- TEST_ABC

Shows:

- Miss
- Cache population
- Prediction fallback

✓ Fast consecutive queries

Shows improved average latency.

✓ Switch dataset MID-DEMO

1. Healthcare → Search
2. Switch → Banking
3. Show predictor reload message
4. Banking queries

📌 Part 6 — What the Examiner Will Notice

They will see:

- AES-Encrypted cache
- RSA-wrapped AES key
- Real-time predictor
- Multidomain behavior
- Dynamic preloading
- Reduced latency
- Hit/Miss variation
- Dataset switching

📌 Part 7 — Shortcut Demo Script (Say This)

**“Here you can see a cache miss. The system fetches fresh data, encrypts it using AES, wraps the key with RSA, and stores it securely.
Now if I repeat the same query, it's a cache hit — decrypted instantly.

This time when I search a scan record, the AI predictor automatically preloads the next likely items based on session patterns. That’s why when I now query this item, it's already a cache hit before I even search it.”**

📌 Part 8 — Everything You Need to Show

✓ Cache Hit
✓ Cache Miss
✓ Prediction Hit
✓ Dataset Switching
✓ AES Encryption
✓ RSA Key Wrapping
✓ Latency Reduction
✓ Preloaded predictions
✓ Domain-agnostic behavior